ANSYS Mechanical

2024

Overview

Problem Specification

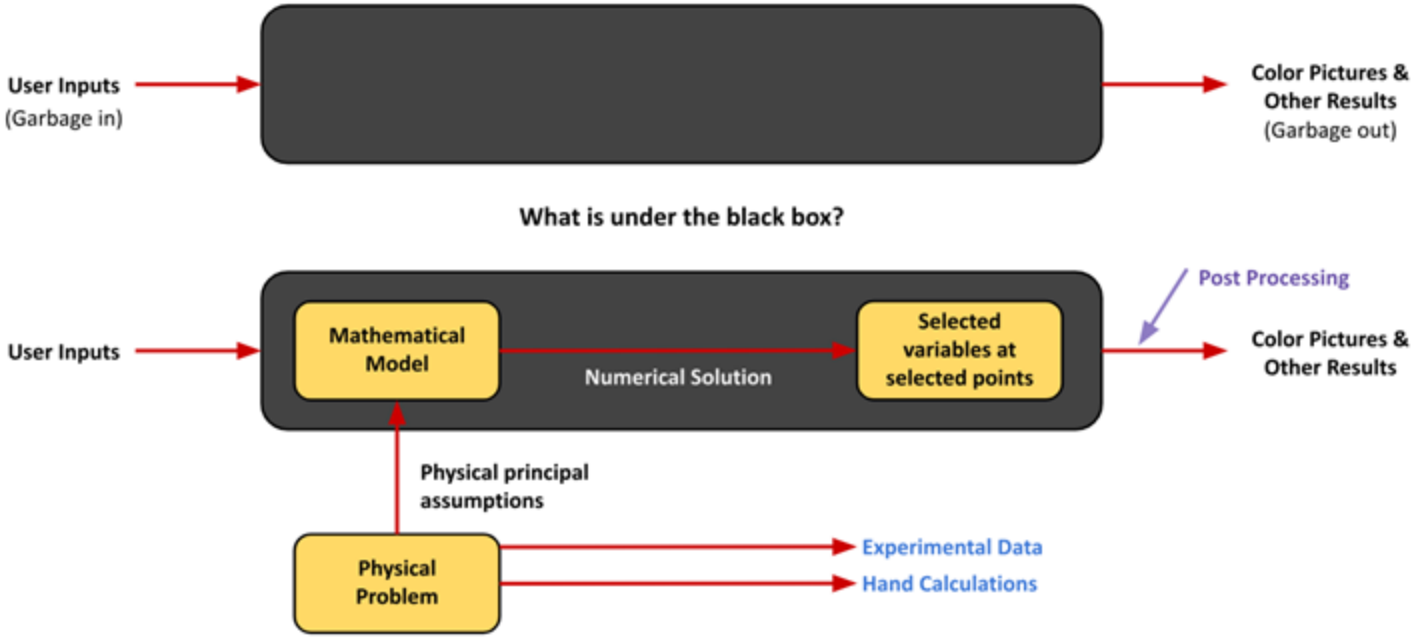
1. Pre Analysis
2. Geometry
3. Model Set-up
4. Connections
5. Mesh
6. Loads and Fixtures
7. Analysis Settings
8. Numerical Solution
9. Verification + Validation

Examples

Cornell EdX:

1. Bike Crank
2. Bolted Nozzle Flange

Black Box


User Inputs
(Garbage in)
Color Pictures &
Other Results
(Garbage out)
What is under the black box?
User Inputs
Mathematical Model
Numerical Solution
Selected variables at selected points
Post Processing
Color Pictures &
Other Results
Physical principal assumptions
Physical Problem
Experimental Data
Hand Calculations

Pre-Analysis

Overview

Mathematical model selection

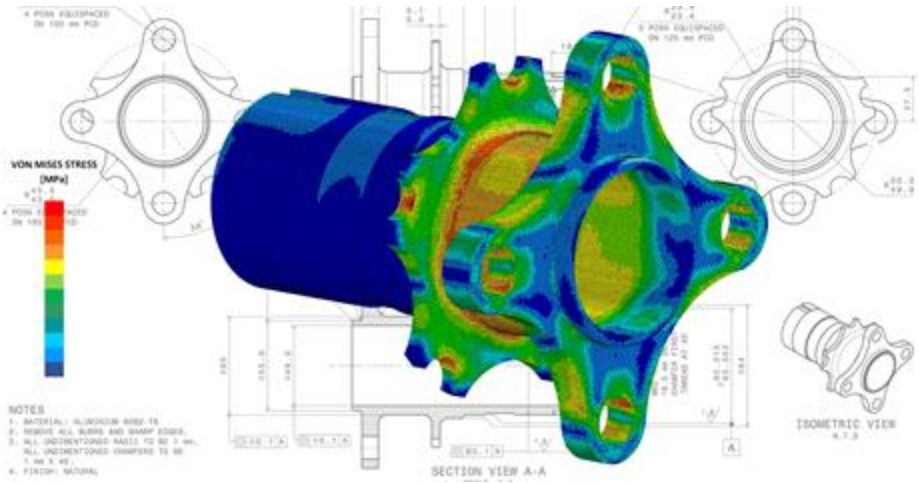
- Physical principles
- Assumptions

Numerical Solution

- Strategy
- Errors introduced
- How to minimize

Hand calculations of expected results or trends

- It doesn't necessarily have to be the area of interest where you do the hand calc, just somewhere you can easily predict results.



Steps

1. Define Governing Equations
    - Define any additional equations
1. Define Boundary Conditions
2. Define Strategy
3. Hand Calculations

Every analysis should be defined with equations and have hand calculations to back up results. There are many models to be used for this, it will not be covered in this presentation.

We do not want button pushers.

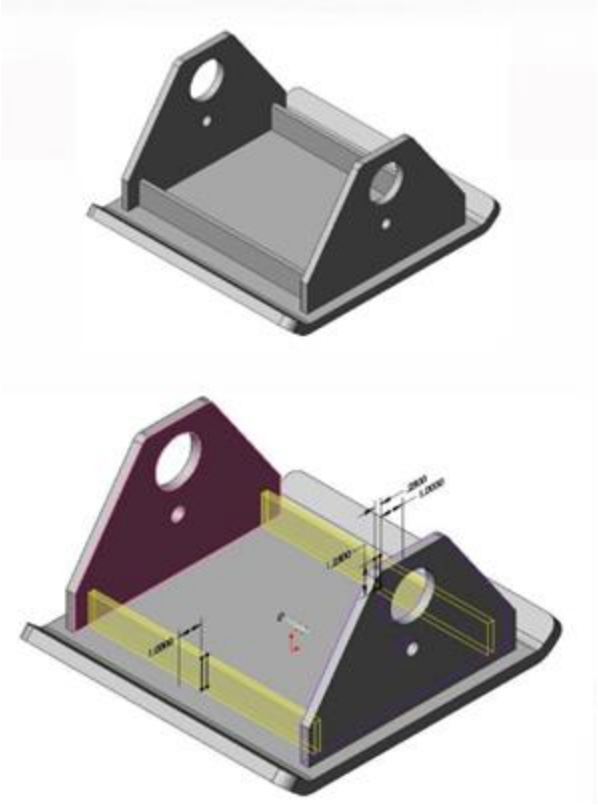
Geometry

File Setup

- Best practice is to create an FEA folder on a local machine or desktop as a location to save all models and results
- Do not run an analysis with the working directory being a network location
- Keep in mind file structure and naming conventions for consistency during iterations
- **Save often**

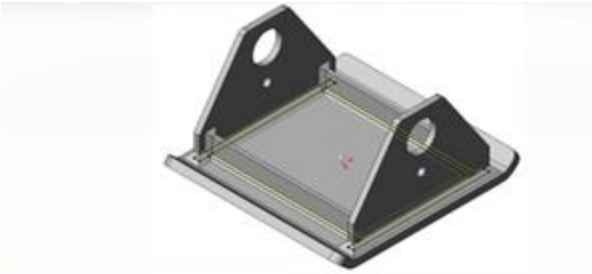
Overview

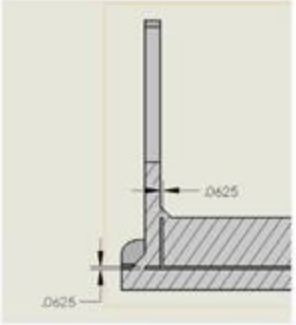
There are a number of techniques used to gain more accurate simulations within the part model



Weldments

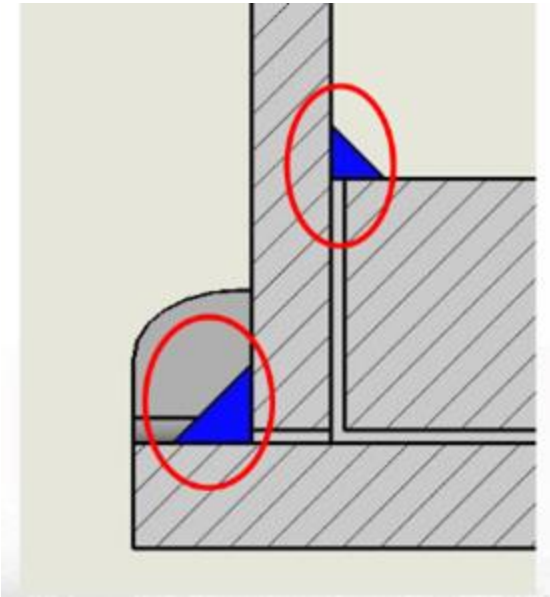
- Weldments are modeled as a single part file
- Within an FEA weldment, components are connected together with welds
  - Fillet welds are modeled as chamfers with leg lengths equal to the weld size
  - Use the Chamfer feature to create fillet welds
- Add gaps between components that contact one another. Gaps should be modeled at .0625 in.
- Without gaps the individual components are connected not only at the welds, but also across the respective plate thickness – not representative of the actual component





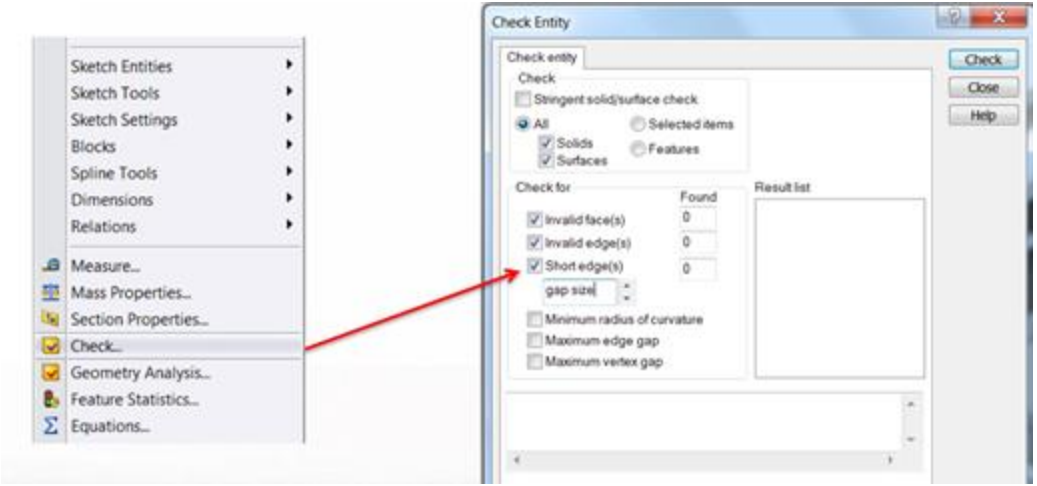
Weldments

- Unless data supports otherwise, welds are modeled with no penetration as a conservative approach
- Once all the features of the weldment have been completed, the model should be one Solid Body



Utilizing Solidworks Features

- Checking for short edges smaller than the minimum gap size in the part, (Tools->Evaluate->Check), will ensure proper aspect ratios after meshing



Utilizing Solidworks Features

For large assemblies that may have many areas of contact, Contact Visualization within a Solidworks Simulation study can be beneficial to confirm part interactions are occurring as expected

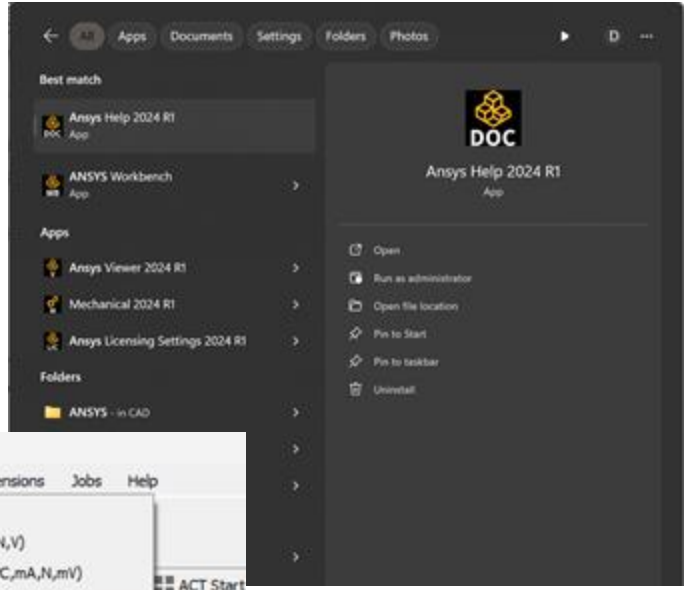
This can also be done with Contact Tool in ANSYS – see slide 32

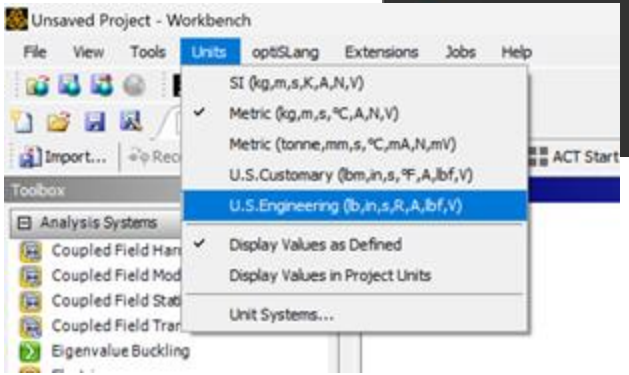
Evaluate -> Interference Detection will ensure parts are not overlapping or interfering with one another

Model Setup

Launching ANSYS Workbench

- Install ANSYS on your machine
- Launch “Workbench XX.X”
- Once Workbench has launched, save project in same location as part or assembly files.
- Select appropriate units for study



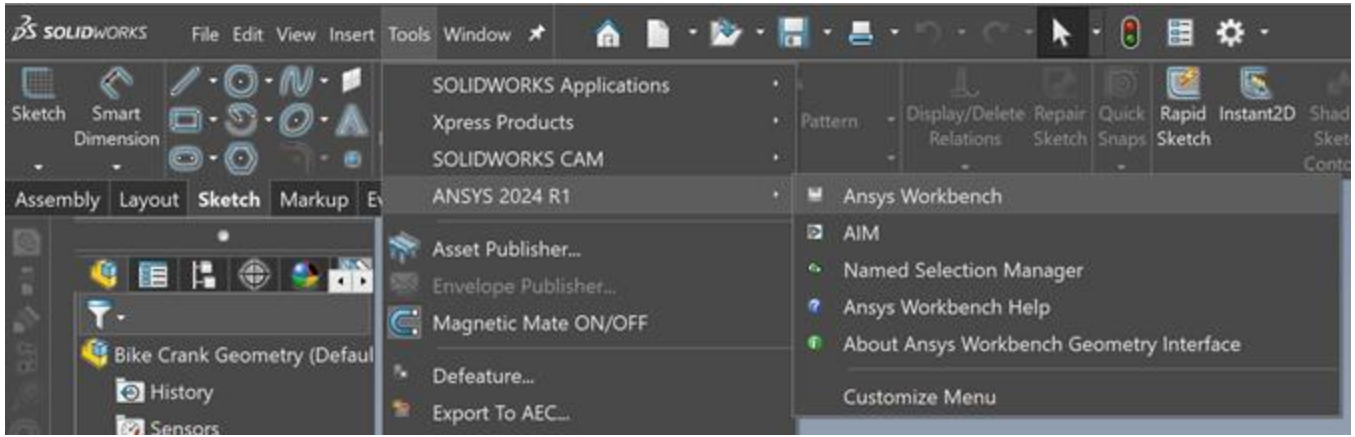


Note: You may have to right click on the taskbar window and press maximize.

Launching ANSYS Workbench

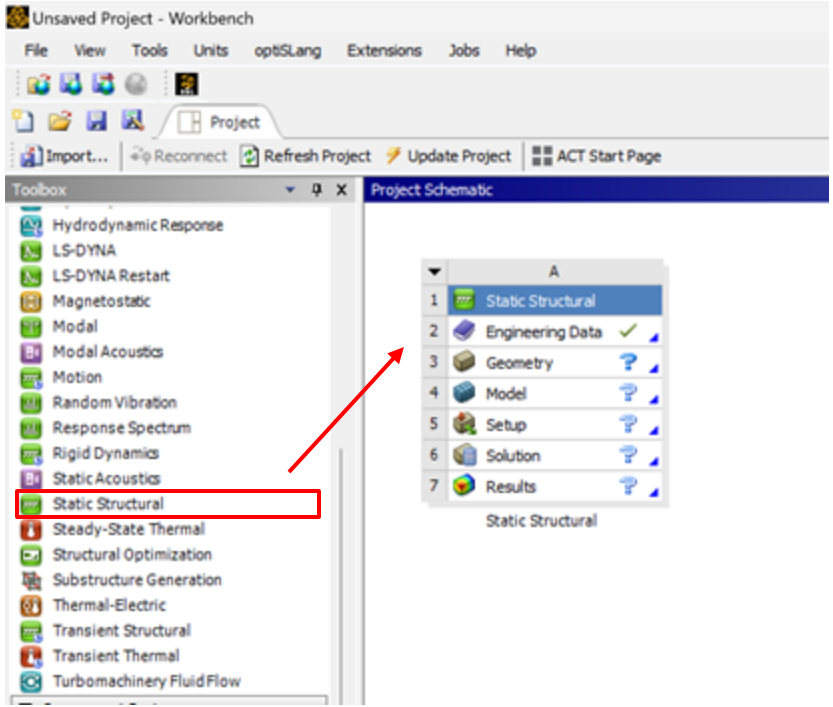
Workbench can also be launched from Solidworks through Tools -> ANSYS XX.X -> ANSYS Workbench.

This will launch Workbench and any part or assembly in the active SW pane will be imported as a Geometry block in the project schematic.



Setting up Project Schematic

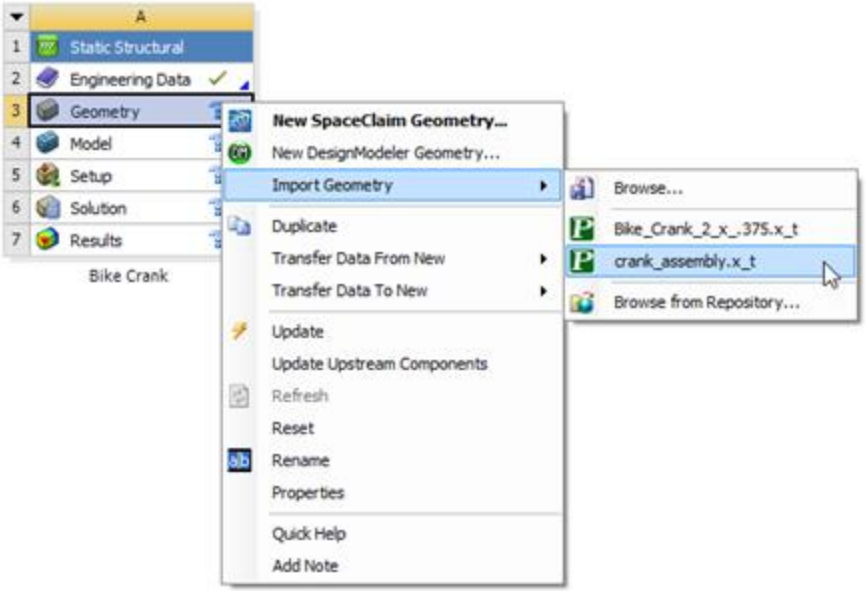
- ANSYS Workbench has a schematic based set up with blocks for each specific task
- Blocks are dragged in from the Toolbox on the left to the Project Schematic on the right
- The Static Structural block has everything needed to run a structural analysis



Adding Geometry

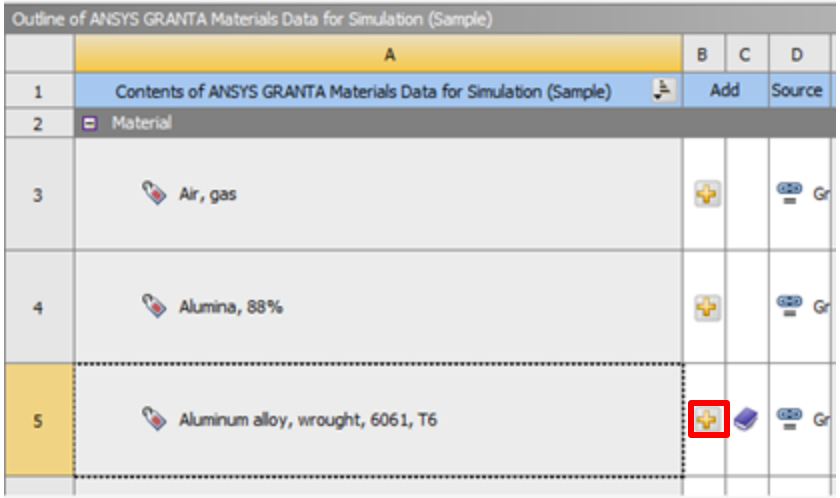
To import the Parasolid file, right-click on the Geometry cell in the Workbench window. Select the parasolid file.

After importing the Parasolid file, right click on the Geometry cell again and select Edit Geometry in DesignModeler. Once the DesignModeler is open, click on Generate and you will see the geometry. Close the DesignModeler window. If you skip this step, your geometry will not be recognized later.



Adding Material Data

- Double click Engineering Data
- Click Engineering Data Sources
  - Click the “+” button to add the material to the schematic
- Click the Engineering Data Sources again to go back
- Close the tab when you are done



Note: watch this video for a visual.

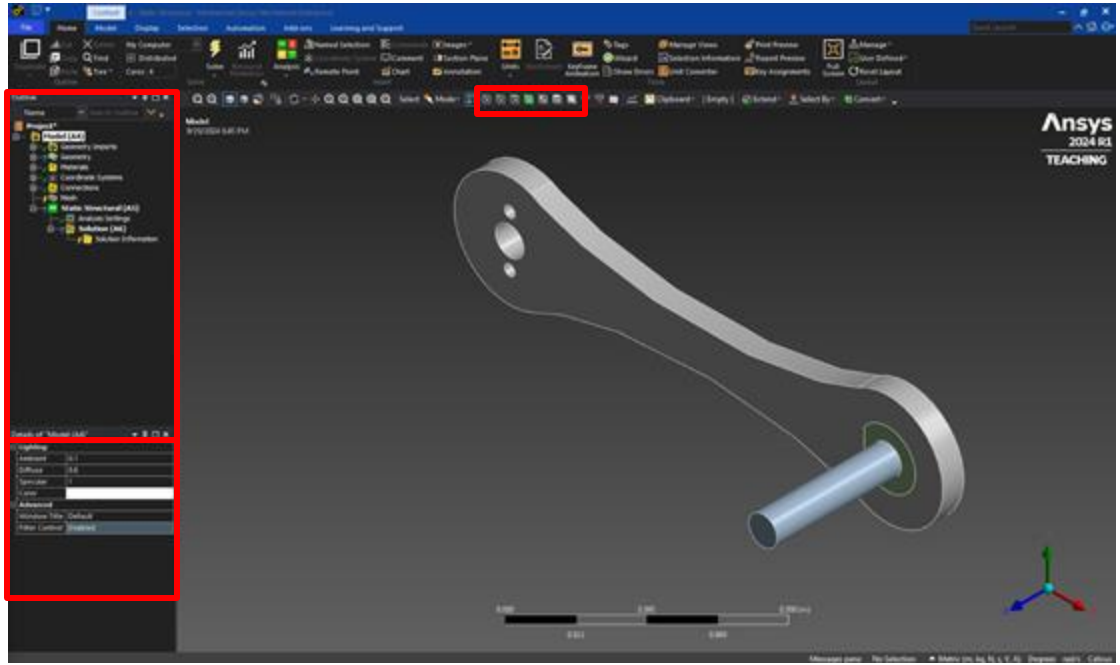
Starting ANSYS Mechanical

- Double click on Model in the Static Structural block to open up ANSYS Mechanical

Selection Filters

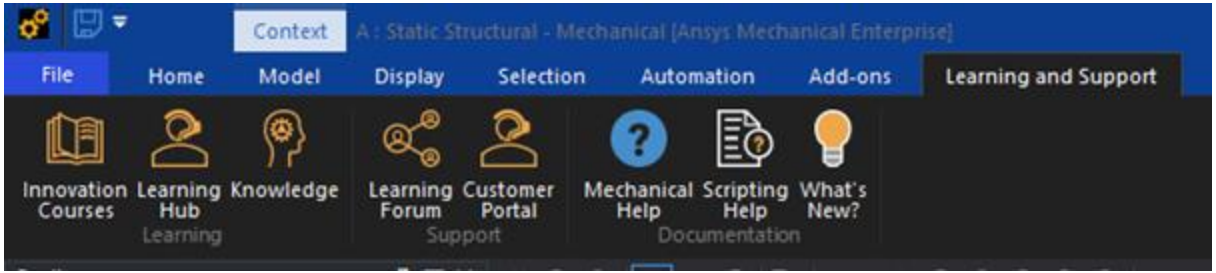
Project Tree

Details and settings of selected item



ANSYS Mechanical Help

- Selecting Mechanical Help will take you to the ANSYS Mechanical online user guide
- This source has detailed explanations for any Mechanical feature



Project Tree Breakdown

Parts/Bodies

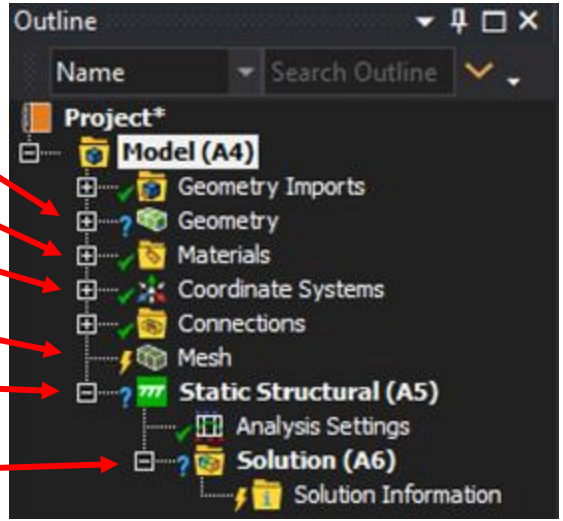
Available Materials

Connections/Contact

Mesh Parameters

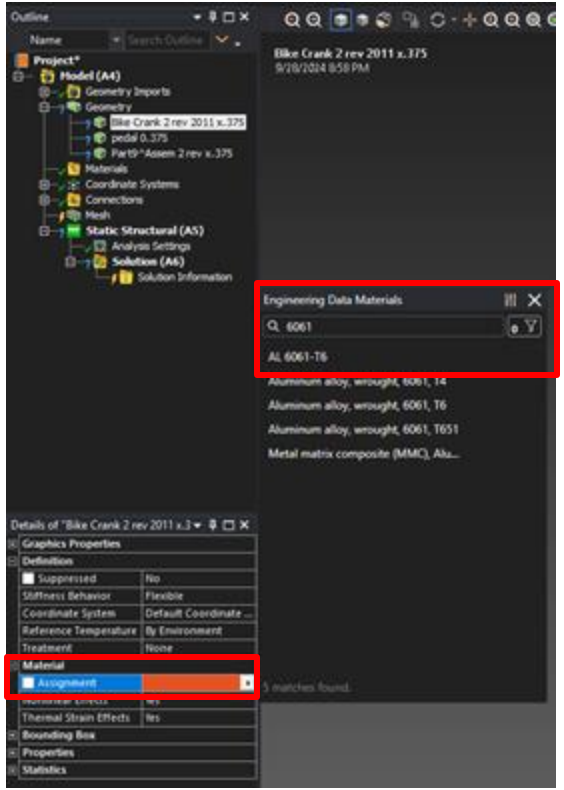
Fixtures/Loads

Solutions/Results



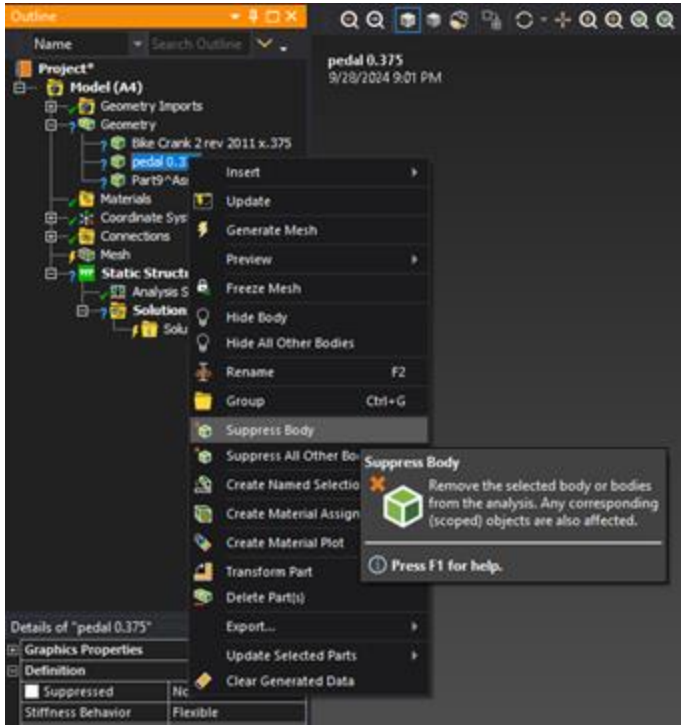
Applying Materials

- Expand Geometry to see all parts and multibody parts
- Define proper material for each part/body in the details box



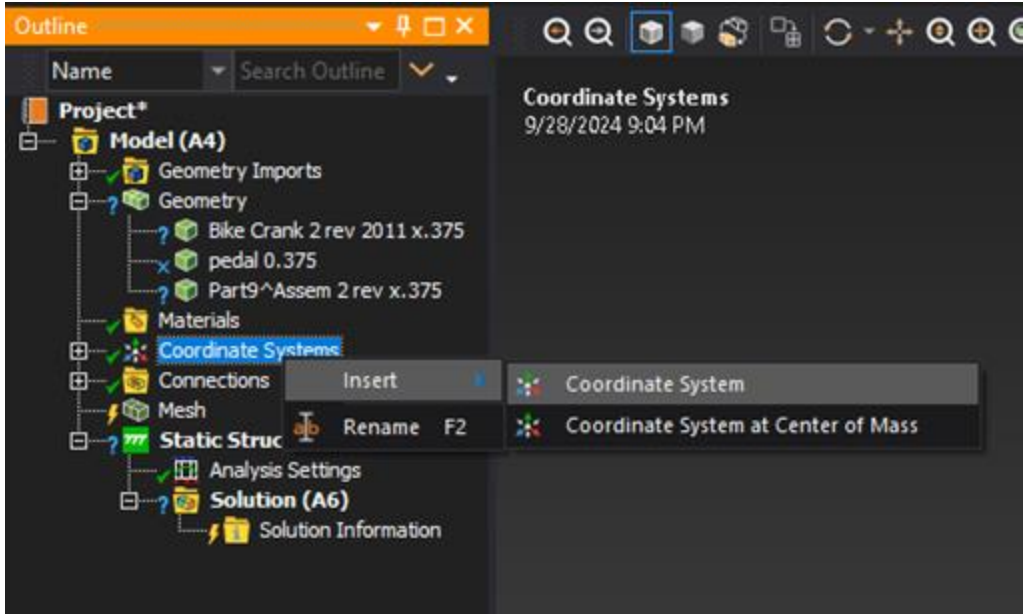
Selecting Geometry

- If a body is to be excluded from the analysis, suppress rather than deleting. If suppressed, body will remain suppressed after updating geometry.



Coordinate System

- If needed, a coordinate system can be created based on geometry or relative to other coordinate systems in the analysis



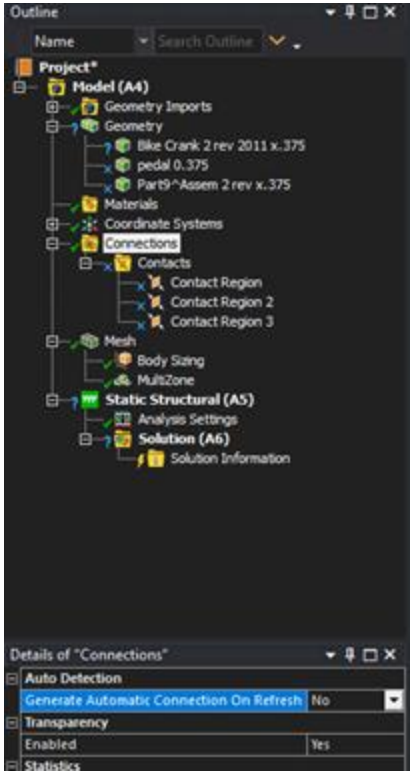
Connections

Connections

Click on Connections, change Generate Automatic Connection On Refresh to No

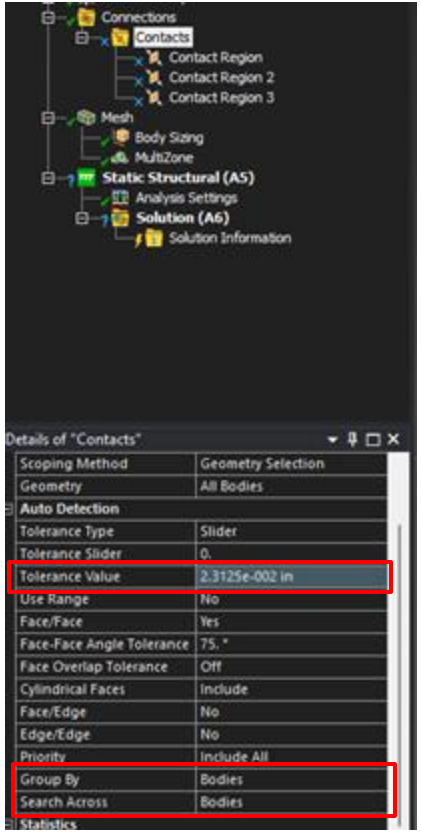
- This keeps your connections from changing after a geometry update

If contact is not needed, suppress rather than delete



Connections

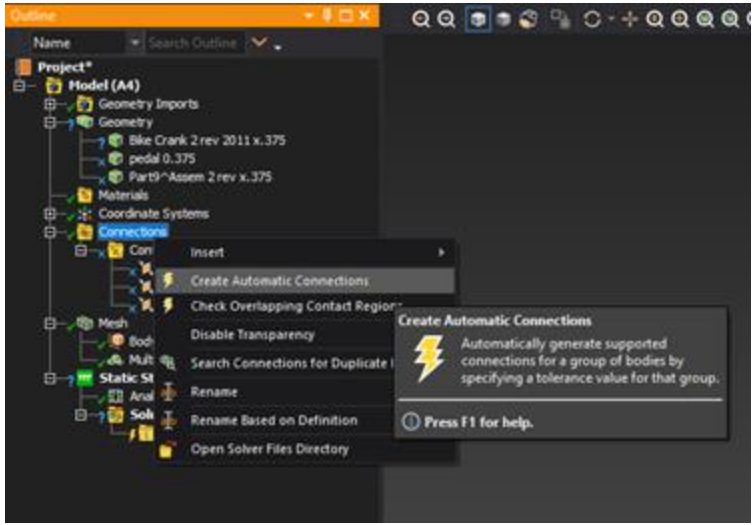
- Click on Contacts, ensure tolerance value is below minimum gap size in model
- Change Group By to None and Search Across to Bodies. These can be changed based on user preference.



Connections

Right click on Contacts -> Create Automatic Connections

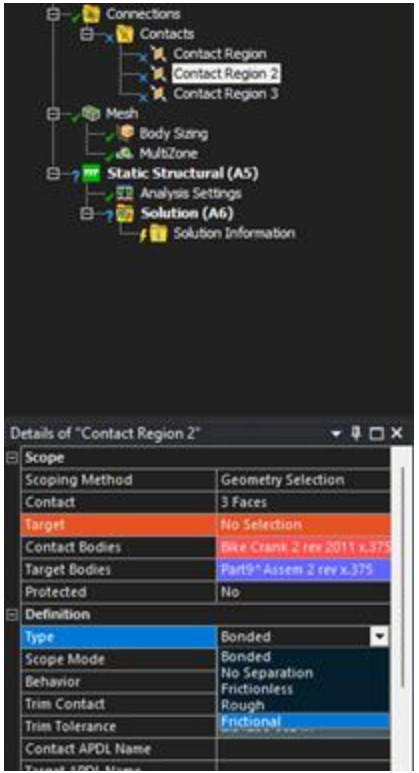
- This will create individual surface contacts between any faces that are touching or within the Tolerance Value



Connections

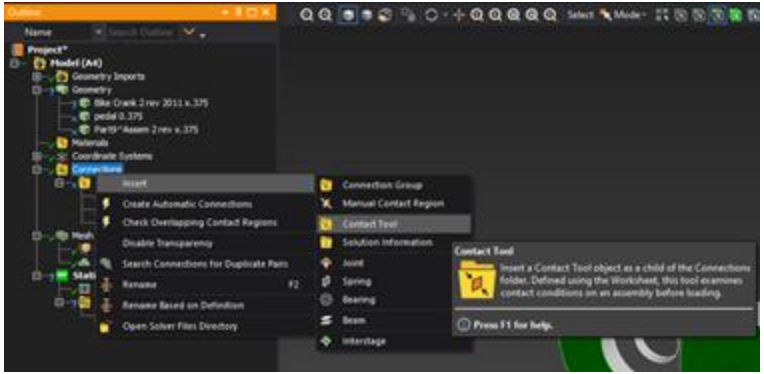
Default contact is bonded, connection type can be changed if bonded is not appropriate

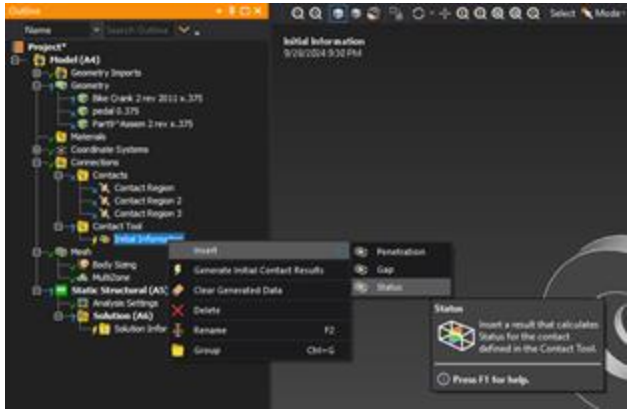
- Frictionless does what No Penetration did in Solidworks



Connections

- Right click on Connections -> Insert -> Contact Tool
- Right click on Initial Information -> Insert -> Status
  - This chart will show contacts similar to “Contact Visualization” in Solidworks

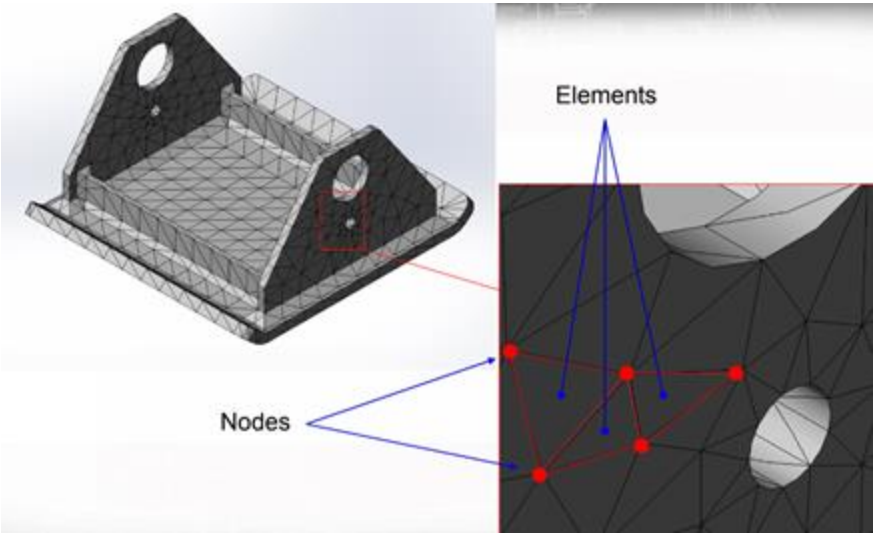




Mesh

What is a “Mesh”?

- The most basic ingredient of a simulation, it is an estimation of the 3D shape of the part.
- A grid of finite points (aka “nodes”) which form the structure of the FEA model.
- Nodes are connected by edges to form distinct 3D chunks, or “elements”



Why is the Mesh important?

- Quality In = Quality Out
- If the mesh is low quality, there is nothing that can be done to give higher quality results
- You spent so much time modeling your part accurately, don’t waste that effort by cutting corners on the mesh

What to know before meshing

- Thickest and thinnest material
- Gap thickness
- Hole diameters vs. material thickness
- Radius/fillet sizes
- *All these could impact the required mesh settings and resultant mesh quality*

Meshing

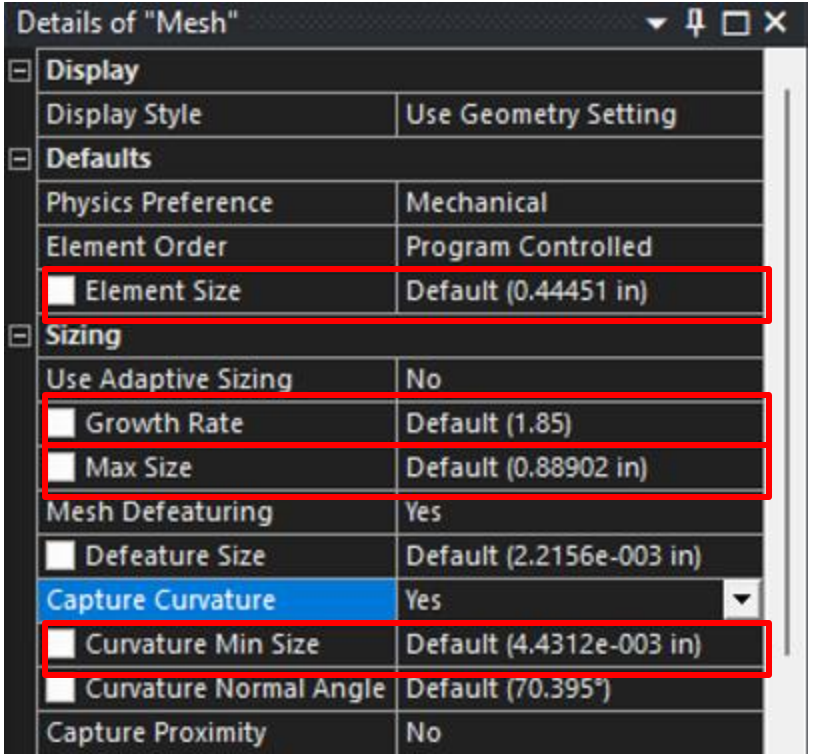
- Click on Mesh
- Change Use Adaptive Sizing to No
- Change Capture Curvature to Yes

Details of "Mesh"

Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	Default (0.44451 in)
Sizing	
Use Adaptive Sizing	No
Growth Rate	Default (1.85)
Max Size	Default (0.88902 in)
Mesh Defeaturing	Yes
Defeature Size	Default (2.2156e-003 in)
Capture Curvature	Yes
Curvature Min Size	Default (4.4312e-003 in)
Curvature Normal Angle	Default (70.395°)
Capture Proximity	No

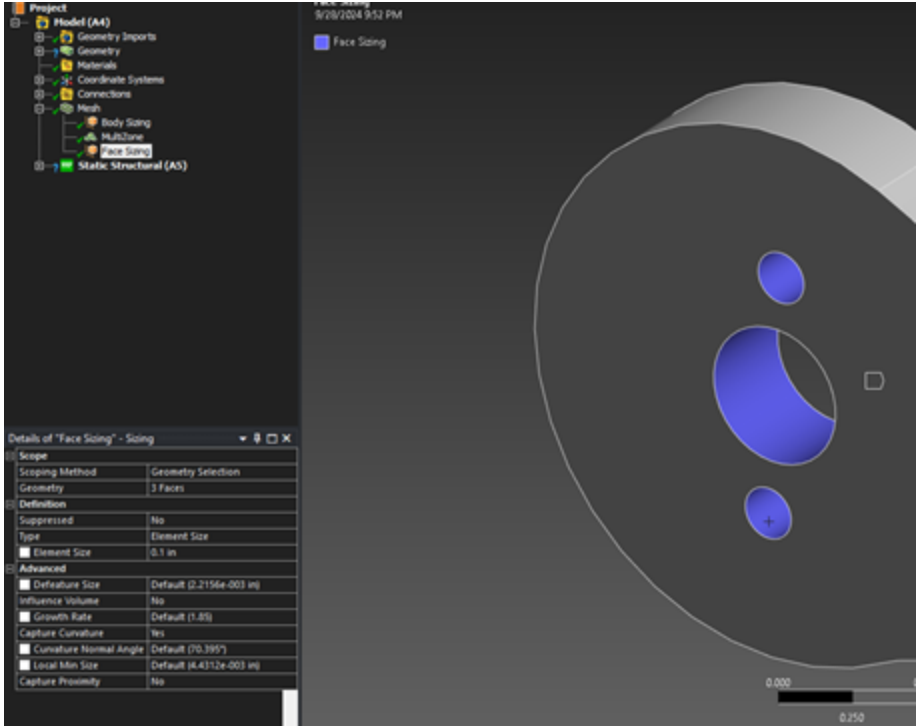
Meshing

- Can alter Element Size, Growth Rate, Max Size, and Curvature Min Size as needed
- Generally, a good starting point for maximum element size is 10 times the smallest entity (usually an edge)
  - **Recommended Values**
    - **Element Size:** 1/16” Typical smallest entity in the model
    - **Growth Rate:** 1.5
    - **Max Size:** 10X minimum element size
    - **Curvature Min Size:**



Refining Mesh Locally

- Mesh Control -> Sizing -> Face Selection Filter -> Select Faces
  - For multiple faces, hold down ctrl
  - Set updated element size
- There should be a minimum of 4 elements across the thickness (the direction it's bending)
- More if there is high stress



Mesh Geometry

Look into aspect ratio of your mesh.

- You never want more than 10 anywhere.
- Aim for below 4.

When you make a mesh, you want to map mesh it and make the bricks uniform

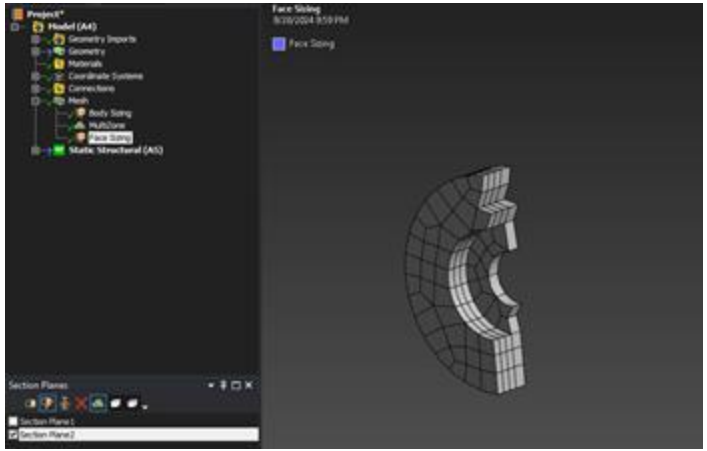
- Use 20-noded bricks most of the time, you can use 8 if it's mostly axial stress, but for bending use 20

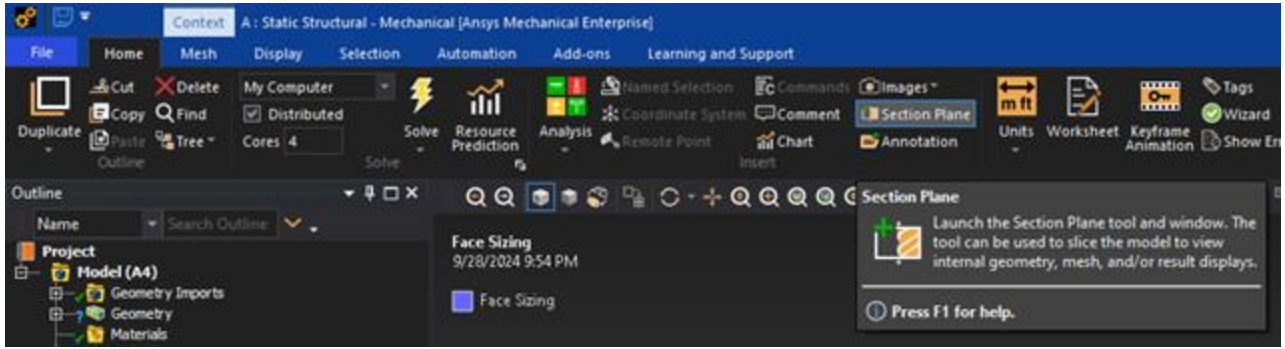
Try not to use prisms or tetrahedrons if you can manage a good mesh.

- They are Stiffer and don't evaluate bending well. If you have a curvy part you might need to hex mesh (use bricks with some others) to get good aspect ratios.

Interrogate Mesh Interior

- Use section planes
- Draw a line over the area you would like to see
- Click, edit section plane to drag along the length of the model

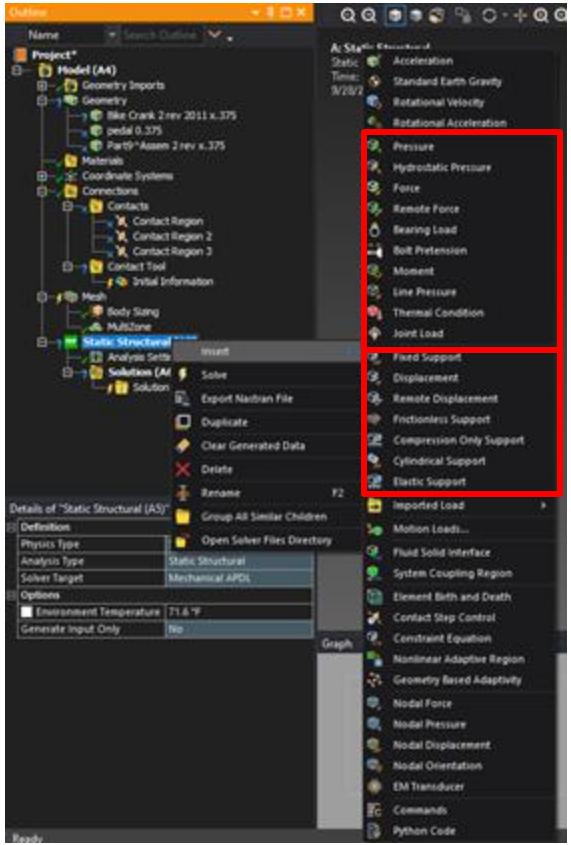




Loads and Fixtures

Loads and Fixtures

- Right click on Static Structural -> Insert to add loads or fixtures
  - Frictionless Support is the same as Roller/Slider in Solidworks
  - Force magnitude is split between all selected faces
- Make sure your boundary conditions are accurate and far away from your area of interest.
- Just understand that if you fix something that actually is a little flexible, there will be a local stress increase nearby.



Analysis Settings

Analysis Settings

Analysis Settings has selections for Solver Type (recommend specifying Direct), Large Deflection, and Time Stepping that may be modified as needed

- Note: turning Large Deflection on forces the model to be non-linear and will require an enterprise license to solve

Details of "Analysis Settings"

Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	
Restart Controls	
Nonlinear Controls	
Advanced	
Output Controls	
Analysis Data Management	

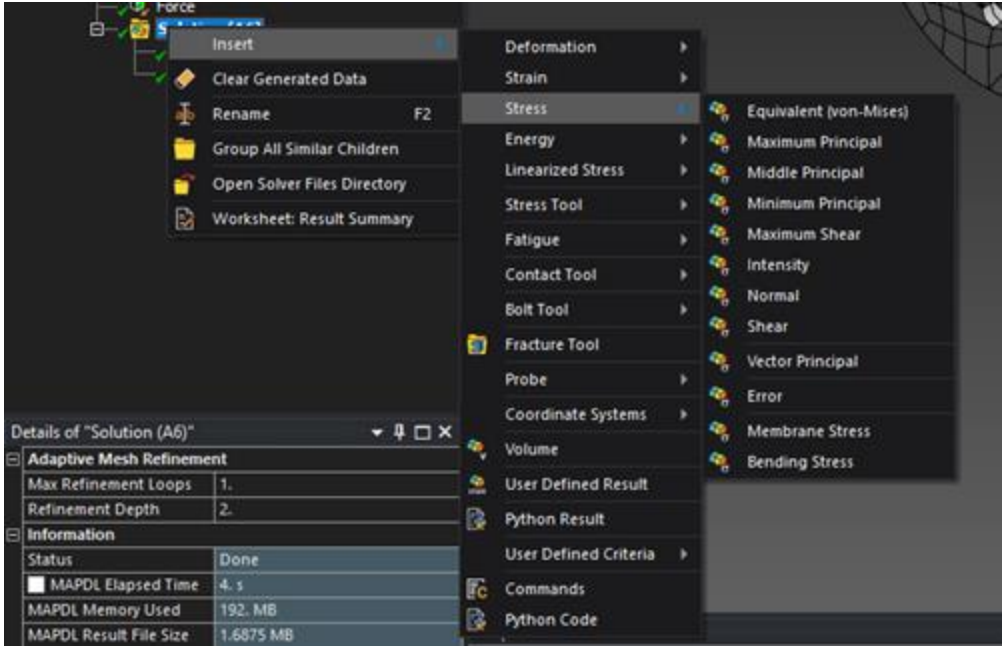
Analysis Settings

- Under Output Controls, change selections for Nodal Forces and Contact Miscellaneous to Yes

Numerical Solution

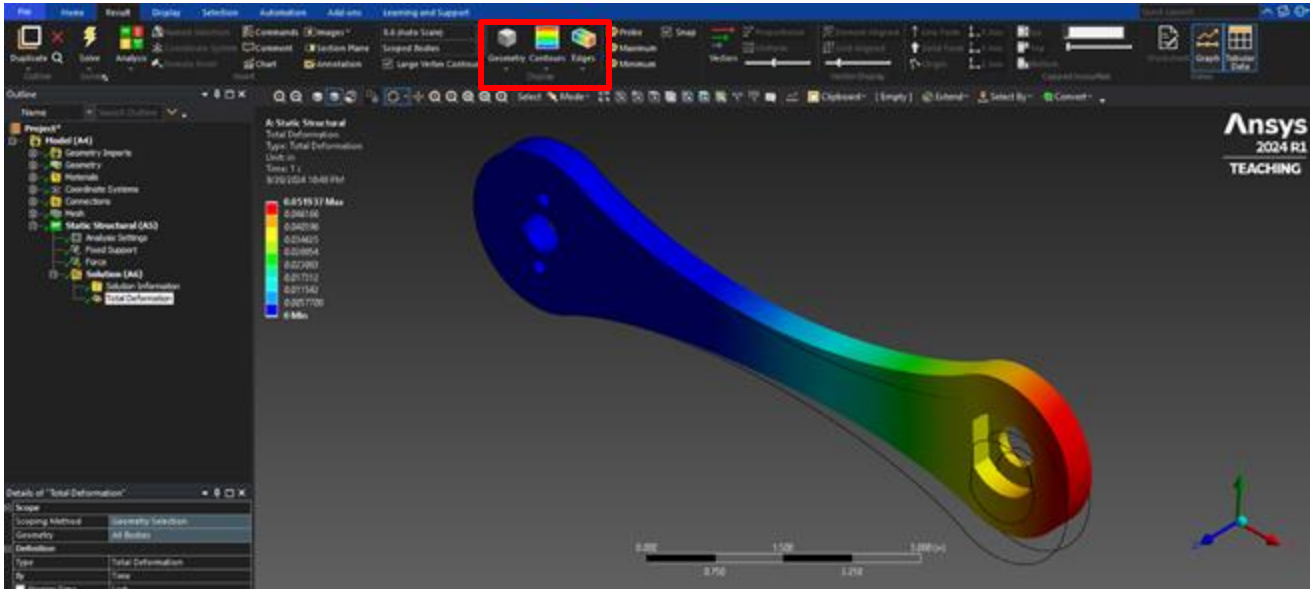
Adding Solution Plots

- Right click on Solution -> Insert to add stress or deformation plots
- Right click on Solution -> Solve to run analysis



Viewing Results

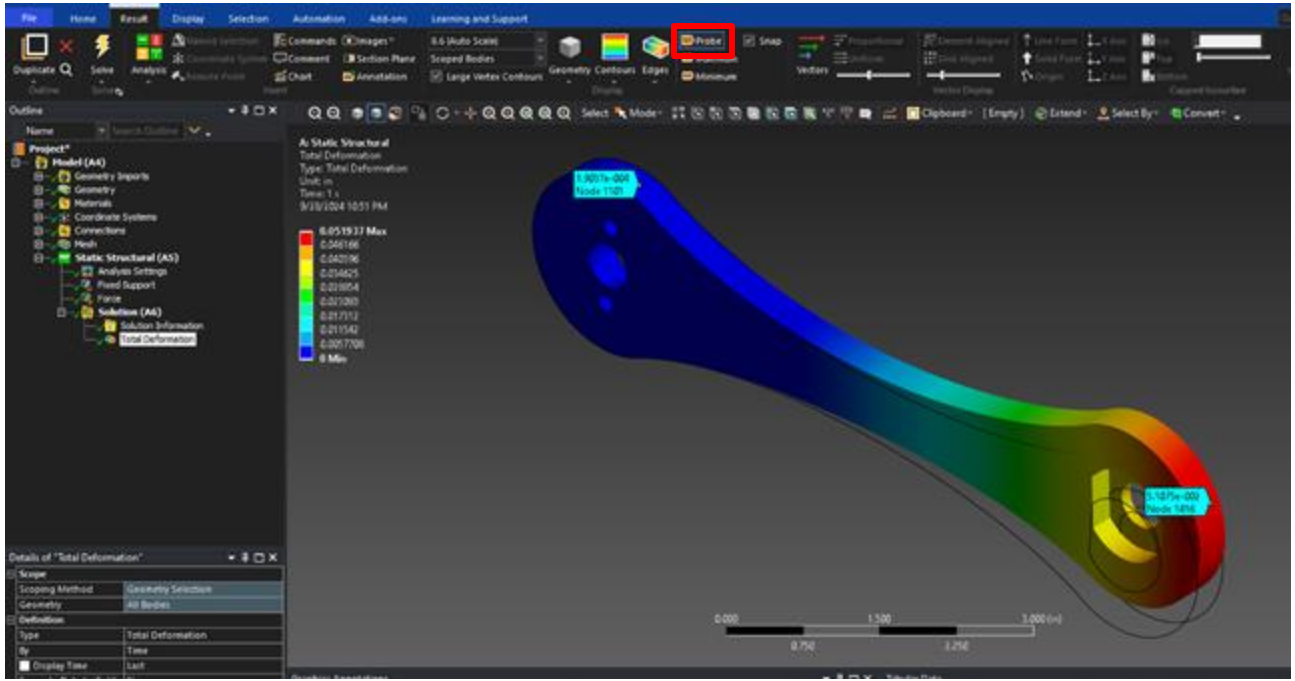
- Various settings for viewing results
- Scale colors and values can be manually adjusted by double clicking on a value or right clicking on the scale itself



Viewing Results

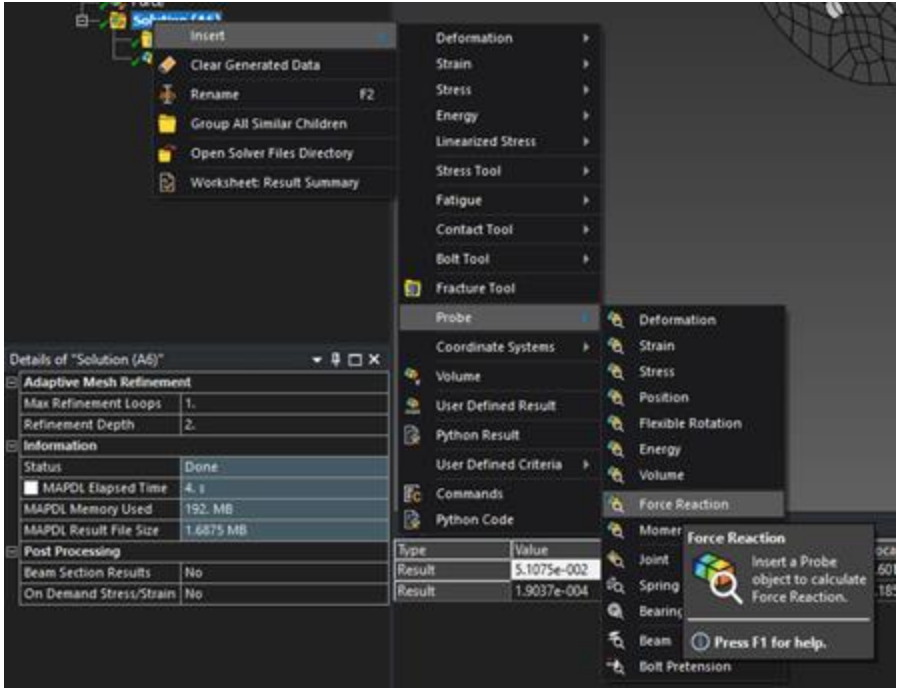
Probe tool adds tags with result value

- Tags can be deleted by clicking the tag selection filter on the top bar, clicking the tag then pushing delete on the keyboard.



Viewing Results

Right click on Solution -> Insert -> Probe -> Force Reaction to get reaction forces



Verification + Validation

Verification + Validation

Check that solution agrees with mathematical model

- Are boundary conditions on displacement and traction satisfied?
- Is equilibrium satisfied?

Check that numerical error is acceptable

- Are ANSYS results reasonably independent of mesh?
- Check comparison with hand calculations for bending stress.

Sensitivity Studies

- The main goal is to ensure that the simulation results are not significantly affected by the size of the mesh elements.
- Process:
  - Run the same simulation using different mesh resolutions.
  - Analyze how the converged solution changes with each mesh.
  - Plot the resulting stress or displacement values for different mesh sizes.
- Helps in identifying the optimal mesh size that provides accurate results without unnecessary computational cost.
- Once you see little variation in the results, you have verified that the mesh is not affecting the validity of your results.

- Video from ANSYS: https://www.youtube.com/watch?v=cm_TtbtD_qM

Sensitivity Studies

To be added later…

End Section 1

Example: Bike Crank

Steps

Problem Specification

1. Pre Analysis
2. Verification + Validation

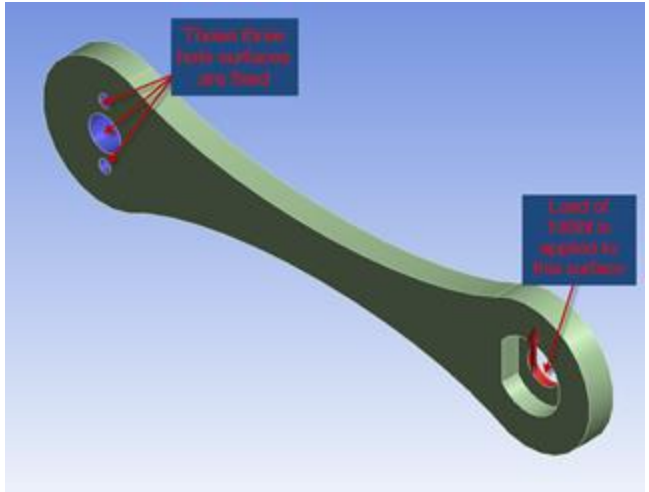
Problem Specification

Consider a crank mounted on a bicycle as illustrated in the picture below. When a rider is pedaling the bicycle, she or he is applying a variable force to this bike crank. To simplify the analysis, we'll neglect the variation of the force with time and focus on the response of the crank to a static force. This is analogous to taking a video of the bike crank and analyzing a single frame by itself.

Problem Specification Cont.

In this module, we will work with the following crank model. The geometry will be provided to you as a download. The crank is made of aluminum 6061-T6 alloy. For this material, the Young's modulus is 1x107 psi and the Poisson's Ratio is 0.33.

The left three hole surfaces are fixed and a load of 100 lbf is applied on the right hole surface in the upward direction. Note that this is an approximation of the actual loads and constraints on the bike crank.

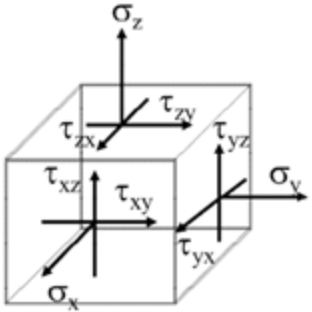


1. Pre-Analysis

Governing Equations:

Physical principle: Equilibrium of infinitesimal element

$\vec{F} = \cancel{m}\vec{a}$ or $\Sigma\vec{F_i} = 0$



3D Differential Equations of Equilibrium

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} + \cancel{f_x} = 0$$

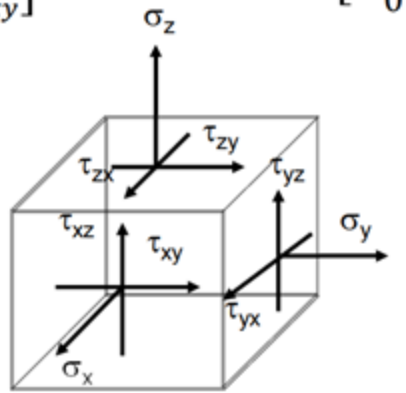
$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} + \cancel{f_y} = 0$$

$$\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} + \cancel{f_z} = 0$$

- 3 eqs.: Force balance in x, y, z directions
- 6 unknown fields: $\sigma_x, \sigma_y, \sigma_z, \tau_{xz}, \tau_{yz}, \tau_{xy}$
- Need additional equations

1. Pre-Analysis Cont.

$$6\begin{bmatrix}\sigma_x\\ \sigma_y\\ \sigma_z\\ \tau_{yz}\\ \tau_{xz}\\ \tau_{xy}\end{bmatrix} = \frac{E}{(1+\nu)(1-2\nu)}\begin{bmatrix}1-\upsilon & \upsilon & \upsilon & 0 & 0 & 0\\ \upsilon & 1-\upsilon & \upsilon & 0 & 0 & 0\\ \upsilon & \upsilon & 1-\upsilon & 0 & 0 & 0\\ 0 & 0 & 0 & 1-2\upsilon & 0 & 0\\ 0 & 0 & 0 & 0 & 1-2\upsilon & 0\\ 0 & 0 & 0 & 0 & 0 & 1-2\upsilon\end{bmatrix}\begin{bmatrix}\varepsilon_x\\ \varepsilon_y\\ \varepsilon_z\\ \gamma_{yz}\\ \gamma_{xz}\\ \gamma_{xy}\end{bmatrix}6$$



Elastic

Isotropic

$$\varepsilon_x = \frac{\partial u}{\partial x} \qquad \varepsilon_y = \frac{\partial v}{\partial y} \qquad \varepsilon_z = \frac{\partial w}{\partial z}$$

Small strains

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \qquad \gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \qquad \gamma_{zx} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}$$

1. Pre-Analysis Cont.

Equations Summary

- 3: Force balance on infinitesimal element in x, y, z directions
- 6: 3D Hooke's law
- 6: Strain-displacement relations

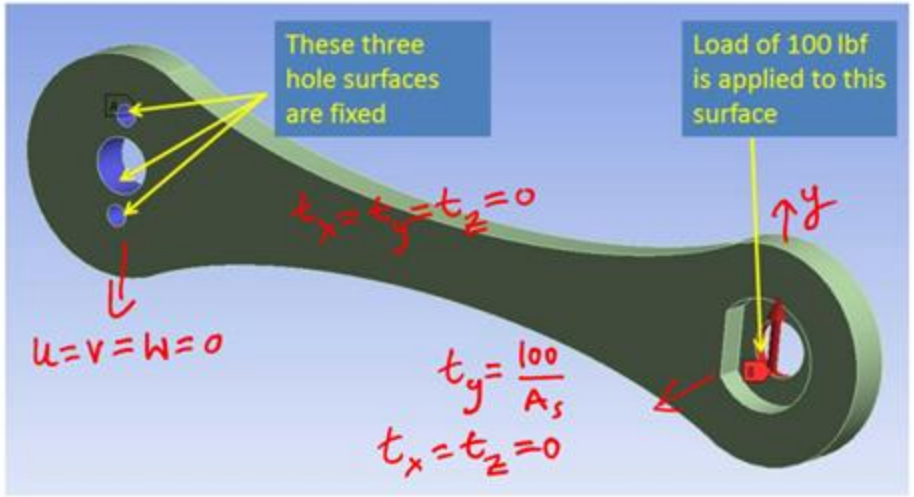
Unknown fields

- 6 stress components
- 6 strain components
- 3 displacement components

1. Pre-Analysis Cont.

At every point on the boundary, the displacement or traction has to be defined.

- Normal as well as tangential directions

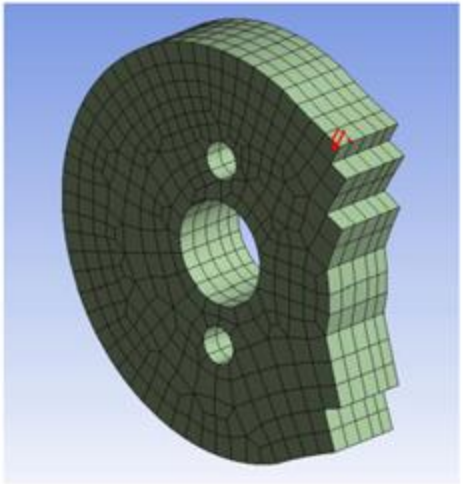


1. Pre-Analysis Cont.

Numerical Solution Strategy

1. Divide domain into elements
2. Reduce the problem to determining displacement values only at nodes.

- Use polynomial interpolation to relate displacements at any point within an element to its nodal displacements.



8 Corners
12 Sides
20 nodes/ele.

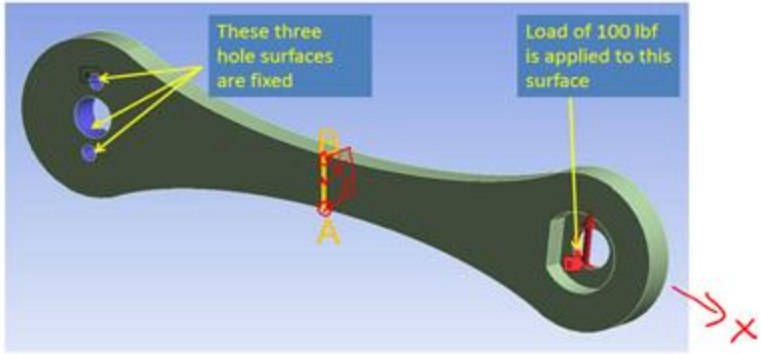
1. Pre-Analysis Cont.

- This loading condition looks like a beam in bending.
- E-B beam theory: For constant cross-section

$$\sigma_x = \frac{My}{I}$$

- Can extend to varying cross- section if rate of variation is "small"

$$\sigma_x = \frac{M\,y(x)}{I(x)}$$



- At A, σ_x = 14,300 psi
- At B, σ_x = −14,300 psi

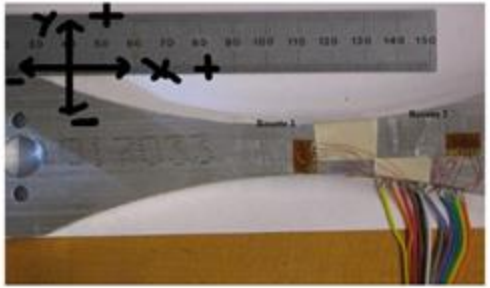
2. Verification + Validation

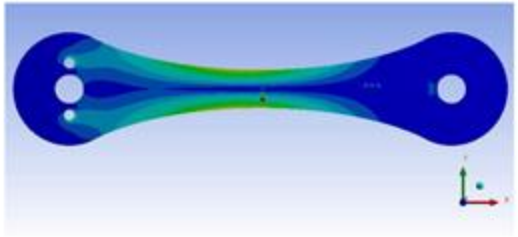
Check that solution agrees with mathematical model

- Are boundary conditions on displacement and traction satisfied?
- Is equilibrium satisfied?

Check that numerical error is acceptable

- Are ANSYS results reasonably independent of mesh?
- Check comparison with hand calculations for bending stress.





Strain gauge	Beam theory	ANSYS	Expt.	% diff.
Rosette 1 Center	317	311	351 ± 6	11

Example:
Bolted Nozzle Flange

Steps

Problem Specification

1. Pre Analysis
2. Verification + Validation